

Supporting Information:

Catalytic competence bounds the decay rates of a thermally formed radical pair

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S1 Software and environment

Spin dynamics: `qbscreen` (this work) with NumPy and SciPy; coherent yields from the Sylvester equation $H_{\text{eff}}X - XH_{\text{eff}}^\dagger = -i\rho_0$ with $H_{\text{eff}} = 2\pi H - \frac{i}{2}(k_S P_S + k_P)$ and $X = \int_0^\infty \rho dt$, solved by the Bartels–Stewart method; [1] yields with electron spin relaxation from the Liouville-space generator. The two agree to 10^{-8} without relaxation, and every yield pair sums to one. Electronic structure: PySCF 2.12.1[2] (B3LYP[3, 4]; def2-SVP for couplings and energies, def2-TZVP[5] for hyperfine couplings; RI-J; PCM), geometries from GFN2-xTB 6.7.1.[6] Every table below is generated from the stored results by `qbscreen/make_si_v6.py`.

S2 The inequality $\Phi_P \leq 4k_P/k_S$ and the dead-end bound

Let $\mathcal{L}$ be the generator of eq 2 of the main text, with $\mathcal{D}$ trace preserving, generating a positivity-preserving evolution (for example a trace-preserving GKSL generator[7, 8]) and unital, $\mathcal{D}(I) = 0$. Then $-\mathcal{L}(I) = k_S P_S + k_P I$. The rest of the generator, $\mathcal{L} - \mathcal{D}$, generates $\rho \mapsto e^{-iH_{\text{eff}}t} \rho e^{iH_{\text{eff}}^\dagger t}$ with $H_{\text{eff}} = 2\pi H - \frac{i}{2}(k_S P_S + k_P I)$, which is positive; by the Lie–Trotter product formula, $e^{\mathcal{L}t} = \lim_{n \rightarrow \infty} (e^{\mathcal{D}t/n} e^{(\mathcal{L} - \mathcal{D})t/n})^n$ is a limit of positive maps and hence positive. For $k_P > 0$ the operator $\mathcal{R} = (-\mathcal{L})^{-1} = \int_0^\infty e^{\mathcal{L}t} dt$ therefore exists and is positive, because the trace of $e^{\mathcal{L}t}(\rho)$ decays at least as $e^{-k_P t}$ for positive ρ . Applying $\mathcal{R}$ to the identity above gives

$$k_S \mathcal{R}(P_S) = I - k_P \mathcal{R}(I) \preceq I, \quad (\text{S1})$$

so $\text{Tr } \mathcal{R}(P_S) \leq 4N/k_S$ with $4N$ the dimension of the spin space, and $\Phi_P = (k_P/N) \text{Tr } \mathcal{R}(P_S) \leq 4k_P/k_S$.

The factor of four cannot be reduced for the class. With $H = 0$, no nuclei, $k_S = 1$ and $k_P = 10^{-6} \mu\text{s}^{-1}$, the fully depolarizing generator $\mathcal{D}(\rho) = \gamma(\text{Tr } \rho I/4 - \rho)$, which is trace preserving and unital, gives $\Phi_P k_S / 4k_P = 0.25$ at $\gamma = 0$ and 0.999996 for $\gamma \geq 10^2 \mu\text{s}^{-1}$ (`competence_sensitivity.sharpness_example`; `calibration_results/phi_bound_sharpness.json`). Trace preservation is not needed for $\Phi_P \leq 4k_P/k_S$ but is needed for $\Phi_P \leq 1$: the unital, completely positive but population-creating generator $\mathcal{D}(\rho) = \gamma(V\rho V^\dagger - \frac{1}{2}\{VV^\dagger, \rho\})$ with $V = |T_0\rangle\langle S|$ gives $\Phi_P = 150/101$ at $k_S = 0.01$, $k_P = \gamma = 1 \mu\text{s}^{-1}$ (`competence_sensitivity.trace_counterexample`).

For a dead-end pair ($k_P = 0$) the resolvent need not exist: without nuclei and with $H = 0$, for example, the 16×16 generator has rank 7, because triplet states that never reach the singlet are stationary. Integrating $\frac{d}{du} e^{\mathcal{L}_0 u}(I) = -k_S e^{\mathcal{L}_0 u}(P_S)$ instead gives

$$k_S \int_0^t e^{\mathcal{L}_0 u}(P_S) du = I - e^{\mathcal{L}_0 t}(I) \preceq I \quad (\text{S2})$$

for all t , since $e^{\mathcal{L}_0 t}(I) \succeq 0$. The population built up from a constant singlet source $k_f[\text{E} \cdot \text{S}]P_S/N$ is therefore at most $4k_f[\text{E} \cdot \text{S}]/k_S = e^{-\Delta/k_B T}[\text{E} \cdot \text{S}]$ at all times.

The function `random_bound_test` of `qbscreen/competence_sensitivity.py` (seed 20261001; inputs and outputs in `calibration_results/phi_bound_test.json`) evaluates $\Phi_P k_S / 4k_P$ for 300 random spin systems: 0–3 nuclei of spin $\frac{1}{2}$ or 1 with couplings of 1–300 MHz, exchange 10^{-1} – 10^4 MHz of

either sign, axial dipolar tensors of -10^{-1} to -10^3 MHz with random axes, random fields up to 0.1 T and rates, and, for about 30 % of the small systems, single-electron dephasing with T_2^e from 1 ns to 10 μ s. The largest value is 0.9986.

S3 The search for F

For each (k_S, k_P) on a grid of 15 by 13 rates, the exchange coupling is scanned on 99 points with sinh spacing (40 MHz scale), symmetric about zero and extending to $\max(4000 \text{ MHz}, 5(k_S + k_P)/2\pi)$, for dipolar scales $s = 0, 0.3, 1$ and 3 and 25 field directions (the pole and eight azimuths over 2π at polar angles of 30, 60 and 90°; the yield is invariant under $B \rightarrow -B$). Each of the three largest local maxima in J is refined twice on 21-point subgrids, and the four best distinct candidates are polished by a Nelder–Mead search in (J, θ, ϕ, s) with s clipped to $[0, 3]$ (160 evaluations each). A cell whose maximum lies within 5 % of the range limit is searched again with a range of sixteen times the position of the maximum. Because the response has narrow resonances in the dipolar scale (at $k_S = 10^4$, $k_P = 10^{-2} \mu\text{s}^{-1}$ and $J \approx 0$ it is 0.40, 0.14, 0.68 and 0.04 % at $s = 0, 0.003, 0.01$ and 0.03 at the first-pass optimum; `calibration_results/s_resonance_example.json`), every cell is then searched a second time: s on 41 values (0 and 40 logarithmically spaced from 10^{-3} to 3), J on 21 points of the same sinh grid plus the first-pass optimum, all 25 directions, followed by a polish of the four best candidates (`worst_case_map.supplement`). The cell keeps the larger value. The second pass raised 151 cells, by up to a factor of 3.6, without changing any row maximum by more than 0.0 %. Repeating it with 121 values of s at the cells that set the thresholds changed F by a factor of at most 1.018 (`calibration_results/s_convergence.json`). Table S1 lists F in percent.

The value G of the main text is computed from these grid values, with the required rates rounded down to grid rates, and is not a bound on the continuous rate region. At $\Delta = 0.46 \text{ eV}$ and $k_{\text{cat}} = 1 \text{ s}^{-1}$, k_{min} equals the grid rate $k_P = 31.6 \mu\text{s}^{-1}$. Nine permitted off-grid values $k_S = 126\text{--}379 \mu\text{s}^{-1}$, with $J = 0$ and $\pm 0.5 \text{ MHz}$, $s = 0$ and all 25 field directions, give at most 0.0924 %, against $G = 0.0965 \%$ (`competence_sensitivity.offgrid_check; calibration_results/g_offgrid_check.json`). This is a restricted search at off-grid points, not a full one.

Table S1: $F(k_S, k_P)$ (%). Rates in μs^{-1} .

$\log k_P \backslash \log k_S$	-2.0	-1.5	-1.0	-0.5	0.0	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0
-2.0	6.3	12	14	13	10	7.5	6	5.2	4.6	3.7	2.3	1.3	1.3	1.3	1.3
-1.5	2.5	6.2	11	14	13	9.6	7.1	5.3	3.8	2.3	1.3	1.3	1.3	1.2	1.2
-1.0	0.85	2.4	6	11	13	12	8.4	5.4	2.7	1.1	1	1.1	1.1	1	1
-0.5	0.26	0.8	2.3	5.5	9.7	11	9	5.2	2	0.83	0.9	0.78	0.63	0.54	0.42
0.0	0.069	0.22	0.66	1.9	4.4	7.2	7.4	4.4	1.6	0.59	0.62	0.63	0.59	0.43	0.31
0.5	0.014	0.044	0.14	0.41	1.1	2.5	3.6	2.7	1.1	0.37	0.34	0.37	0.35	0.26	0.15
1.0	0.0014	0.0044	0.014	0.043	0.13	0.34	0.66	0.75	0.48	0.2	0.07	0.084	0.088	0.077	0.045
1.5	5.8e-5	1.8e-4	5.8e-4	0.0018	0.0056	0.017	0.043	0.083	0.096	0.062	0.032	0.049	0.055	0.042	0.02
2.0	1.7e-6	5.4e-6	1.7e-5	5.4e-5	1.7e-4	5.3e-4	0.0015	0.004	0.0077	0.014	0.018	0.02	0.023	0.015	0.0043
2.5	1.2e-7	3.8e-7	1.2e-6	3.8e-6	1.2e-5	3.7e-5	1.2e-4	3.5e-4	9.6e-4	0.0021	0.0033	0.0039	0.0036	0.0014	2.6e-4
3.0	4.8e-9	1.5e-8	4.8e-8	1.5e-7	4.8e-7	1.5e-6	4.8e-6	1.5e-5	4.4e-5	1.1e-4	2.3e-4	2.6e-4	1.4e-4	3.3e-5	4.5e-6
3.5	1.0e-10	3.2e-10	1.0e-9	3.2e-9	1.0e-8	3.2e-8	1.0e-7	3.1e-7	9.5e-7	2.6e-6	5.3e-6	5.4e-6	2.2e-6	3.0e-7	5.7e-8
4.0	1.1e-12	2.6e-12	7.6e-12	2.3e-11	7.4e-11	2.3e-10	7.4e-10	2.3e-9	7.2e-9	2.2e-8	5.8e-8	1.2e-7	1.1e-7	3.3e-8	3.2e-9

S4 Dependence on the spin model

In the anisotropic model “aniso” described below, the full search of section S3 (first pass, edge rule and fine- s pass) was repeated on every cell with $k_P = 10\text{--}10^{2.5} \mu\text{s}^{-1}$ and $k_S \geq k_P$ (`model_sensitivity.run_aniso_full; calibration_results/worst_case_v6_aniso.json`); Table S2 lists the values, 0 of which lie at the edge of the J range. The thresholds in the main text replace each of these cells by the larger of its reference and anisotropic values.

The other models were examined only locally: for each onward rate that sets the thresholds of G , the cell that gives the row maximum of F in the reference model was searched again (`qbscreen/model_sensitivity.py`), starting from the reference optimum and a 39-point J grid at its field direction

and dipolar scale, and polishes the two best candidates. In “aniso” N5 and N10 carry the literature isotropic couplings plus the density-functional spin-dipolar tensors (Table S6), rotated into the frame of the contact complex in which the dipolar tensor was computed (0.09 Å root-mean-square deviation between the relaxed anion and the complex flavin). The complex was built with methylamine, so the benzylaminium tensors have no orientation in it: the aminium α -protons are kept isotropic (their anisotropy is below 5 % of the isotropic part), and in “ $+^{14}\text{N}$ ” the aminium nitrogen (spin 1), whose tensor is nearly axial about the singly occupied orbital, is added with its axis along the lone pair of the placed amine. “+2H” adds the two largest flavin methyl protons. The relaxation test uses a reduced pair (N5 isotropic, spin 1; aminium proton 256.9 MHz) searched with the full procedure, without relaxation and with single-electron dephasing (Table S3); the reduced-pair relaxation test is superseded by the direct four-nucleus calculation of section S5 and is shown for comparison. Because only the reference row-maximum cell was searched, and only near its optimum, another cell of the same row may give a larger value; for the anisotropic model the full search above supersedes these entries.

Table S2: $F(k_S, k_P)$ (%) in the anisotropic model from the full search. Rates in μs^{-1} .

$\log k_P \backslash \log k_S$	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0
1.0	1.7	2.3	1.8	0.99	0.52	0.28	0.18	0.14	0.067
1.5		0.23	0.28	0.21	0.13	0.068	0.026	0.017	0.0082
2.0			0.022	0.027	0.021	0.011	0.0049	0.0019	5.1e-4
2.5				0.0021	0.003	0.0025	0.0014	3.7e-4	6.4e-5

Table S3: Searched F (%) at the threshold-setting cells in the reference and richer models, and in the reduced pair with and without electron dephasing.

$\log k_P$	$\log k_S$	reference	aniso	$+^{14}\text{N}$	+2H	red. coh.	10 ns	100 ns	1 μs	10 μs
1.0	1.5	0.75	2.3	1	1.6	0.51	0.21	0.22	0.41	0.5
1.5	2.0	0.096	0.28	0.16	0.2	0.086	0.052	0.038	0.076	0.085
2.0	4.0	0.023	0.0035	0.0033	0.0036	0.0033	0.016	0.0028	0.0032	0.0032
2.5	3.5	0.0039	0.0023	0.0015	0.0022	7.8e-4	0.0012	7.6e-4	7.8e-4	7.8e-4

S5 Electron spin relaxation in the four-nucleus model

The four-nucleus spin space has dimension 144, too large for the dense Liouville-space solution. With $\gamma = 2/T_2^e$, single-electron dephasing $\gamma \sum_e (S_{ez} \rho S_{ez} - \rho/4)$ is split into a decay $-(\gamma/2)\rho$, absorbed into $K = 2\pi H - \frac{i}{2}[k_S P_S + (k_P + \gamma/2)I]$, and a recycling term $\gamma \sum_e S_{ez} \rho S_{ez}$. The integral $X = \int_0^\infty \rho dt$ then solves $KX - XK^\dagger = -i\rho_0 - i\gamma \sum_e S_{ez} X S_{ez}$, which is iterated by GMRES[9] with the left side inverted by a triangular Sylvester solve in the Schur basis of K (product_yield.yields_t2e). On twelve random systems small enough for the dense solution, including anisotropic hyperfine tensors, the two agree to a relative 1.3×10^{-13} . For every cell with $k_S \geq k_P$ on the threshold rows, the first-pass search of section S3 was repeated with $T_2^e = 10$ ns, 100 ns and 1 μs , with the same edge rule for the J range (Table S4; calibration_results/worst_case_v6_t2e.json). Nineteen searches (14 at $T_2^e = 10$ ns, 5 at 100 ns) first peaked at the edge of the J range and were repeated with the range widened sixteenfold; one of them ($k_S = 10^{4.5}$, $k_P = 10 \mu\text{s}^{-1}$, $T_2^e = 100$ ns) still peaks at the edge because the response approaches a plateau as $|J| \rightarrow \infty$: it is constant to five figures for $|J| = 10^6$ – 10^8 MHz and equal to the stored value (calibration_results/t2e_plateau_check.json).

S6 Hyperfine couplings

Fermi-contact couplings (B3LYP/def2-TZVP at GFN2-xTB geometries), all nuclei with $|a_{\text{iso}}| \geq 1$ MHz (Table S5), and the spin-dipolar tensors of the nitrogens and the largest protons (Table S6), computed

Table S4: Searched F (%) without and with single-electron dephasing in the four-nucleus model. Rates in μs^{-1} .

$\log k_P$	$\log k_S$	coherent	10 ns	100 ns	1000 ns
1.0	1.0	0.66	0.075	0.22	0.54
1.0	1.5	0.75	0.12	0.29	0.63
1.0	2.0	0.48	0.18	0.26	0.41
1.0	2.5	0.2	0.21	0.18	0.18
1.0	3.0	0.07	0.22	0.21	0.067
1.0	3.5	0.084	0.22	0.21	0.073
1.0	4.0	0.088	0.22	0.21	0.076
1.0	4.5	0.077	0.22	0.21	0.065
1.0	5.0	0.045	0.22	0.21	0.04
1.5	1.5	0.083	0.024	0.046	0.077
1.5	2.0	0.096	0.042	0.059	0.09
1.5	2.5	0.062	0.063	0.045	0.059
1.5	3.0	0.032	0.072	0.028	0.03
1.5	3.5	0.049	0.075	0.035	0.047
1.5	4.0	0.055	0.076	0.04	0.053
1.5	4.5	0.042	0.077	0.033	0.041
1.5	5.0	0.02	0.077	0.015	0.019
2.0	2.0	0.0077	0.0067	0.0068	0.0076
2.0	2.5	0.014	0.011	0.01	0.014
2.0	3.0	0.018	0.013	0.014	0.017
2.0	3.5	0.02	0.015	0.016	0.019
2.0	4.0	0.023	0.015	0.018	0.022
2.0	4.5	0.015	0.015	0.012	0.014
2.0	5.0	0.0043	0.015	0.0033	0.0042
2.5	2.5	0.0021	0.0016	0.002	0.0021
2.5	3.0	0.0033	0.0024	0.0031	0.0032
2.5	3.5	0.0039	0.0023	0.0036	0.0039
2.5	4.0	0.0036	0.0017	0.0031	0.0035
2.5	4.5	0.0014	0.0013	0.0012	0.0013
2.5	5.0	2.6e-4	0.0012	2.2e-4	2.5e-4

from the spin density with the operator-derivative integrals $\langle \mu | \partial_a \partial_b | \mathbf{r} - \mathbf{R} |^{-1} | \nu \rangle$; their trace reproduces $-4\pi\rho_s(\mathbf{R})$. The computed flavin N5 and N10 isotropic couplings are within 16 % and 13 % of the literature values used in the main text,[10] and the N5 tensor is nearly axial with its large component normal to the ring. For 3-methylindole $^{\bullet+}$ the two large methyl couplings stand in for the tryptophan β -protons; their values depend on the side-chain conformation.

S7 Dipolar tensor

The tensor is the point-dipole tensor summed over pairs of atoms weighted by their Löwdin[11] spin populations (UKS B3LYP/def2-SVP on each radical at its position in the complex), for methylamine on the flavin si face with N5–N = 3.5 Å (Table S7).

S8 Electronic coupling

Fragment-orbital couplings[12] $t = \langle \text{LUMO}_{\text{Fl}} | F | \text{HOMO}_{\text{am}} \rangle$ (B3LYP/def2-SVP) for methylamine and benzylamine stacked on the flavin si face (five orientations) or approaching its edge (three orientations), at N5–N distances of 3.0–6.0 Å. Geometries with any intermolecular contact below 2.0 Å are excluded (Table S8). The edge approach of benzylamine clashes at every distance below 6 Å and is not used.

Table S5: Isotropic hyperfine couplings.

radical	nucleus (atom index)	a_{iso} (MHz)	a_{iso} (μT)
lumiflavin $\bullet^-$	H29	+17.9	+639
	H30	+17.8	+636
	N4	+12.4	+441
	H25	+11.3	+403
	H26	+11.3	+402
	H22	-10.5	-374
	H20	-5.3	-190
	H21	-5.3	-190
	N13	+4.6	+165
	H27	+1.4	+50
	N8	-1.0	-36
methylaminium $\bullet^+$	H2	+363.1	+12955
	H6	+82.7	+2951
	H5	+81.9	+2923
	H3	-59.3	-2116
	H4	-59.3	-2115
	N1	+30.5	+1090
benzylaminium $\bullet^+$	H10	+256.9	+9167
	H11	+196.2	+7001
	N0	+24.8	+885
	H8	-23.0	-821
	H9	-20.7	-737
	H14	-12.3	-439
	H16	-7.8	-278
	H13	-3.2	-115
	H15	+2.4	+85
	H12	+60.1	+2143
3-methylindole $\bullet^+$	H11	+55.5	+1980
	H18	-14.3	-511
	H13	-13.1	-467
	H14	-13.1	-466
	H16	-9.8	-351
	H15	-4.9	-175
	N3	+4.8	+172
	H17	+2.7	+95

S9 Density-functional driving force

The main text uses measured potentials. For comparison, Table S9 gives adiabatic one-electron potentials in water ($\varepsilon = 78.4$, absolute SHE potential 4.44 V[13]) from the same density-functional protocol, against the measured values for a dialkylamine[13] and for free FMN at pH 7.[14] The flavin comparison is between different species (lumiflavin anion radical versus the pH-7 FMN oxidized/semiquinone couple) and indicates only the order of the error. Table S10 gives the raw driving force in low-dielectric continua with a contact Coulomb term $-e^2/(4\pi\varepsilon_0\varepsilon r)$. No zero-point or thermal corrections are included, and none of these values enters the main text.

S10 The driving force in the active site

The main text places the pair in water at $\Delta_w = E_{\text{amine}}^\circ - E_{\text{fl}} - e^2/(4\pi\varepsilon_0 \cdot 78.4r)$, $r = 3.5\text{--}6$ Å. Moving the pair into a medium of effective dielectric constant ε changes the Coulomb term by $-e^2/(4\pi\varepsilon_0r)(1/\varepsilon - 1/78.4)$ and the solvation of the ions, which is not known. Two continuum limits are evaluated for $\varepsilon = 4$ and 10.

One-sided. Only the aminium radical cation is desolvated, with a Born term $e^2/(8\pi\varepsilon_0a)(1/\varepsilon - 1/78.4)$; the flavin anion keeps its aqueous solvation. The radius $a = 2.81$ Å reproduces, by the Born

Table S6: Spin-dipolar hyperfine tensors.

radical	nucleus	a_{iso} (MHz)	principal values of A_{dip} (MHz)
lumiflavin $\bullet^-$	H29	+17.9	-1.2, -1.0, +2.2
	H30	+17.8	-1.2, -1.0, +2.2
	N4	+12.4	-17.0, -16.7, +33.7
	H25	+11.3	-1.5, -0.9, +2.4
	H26	+11.3	-1.5, -0.9, +2.4
	N13	+4.6	-6.0, -5.7, +11.7
	N8	-1.0	-0.2, +0.1, +0.1
	N11	-0.2	-0.6, -0.4, +1.1
	H10	+256.9	-8.0, -4.2, +12.2
benzylaminium $\bullet^+$	H11	+196.2	-7.9, -4.2, +12.1
	N0	+24.8	-18.8, -18.6, +37.4
	H8	-23.0	-21.8, -6.0, +27.8
	H9	-20.7	-21.4, -5.4, +26.8

Table S7: Dipolar coupling of the contact pair.

model	D (MHz)	E (MHz)	principal values of T (MHz)
spin-population tensor	-843	-26	-1124, 511, 613
point dipole at spin centroids	-1642	0	–
point dipole at N5 and N	-1817	0	–

formula in water, the solvation-energy difference of 244 kJ mol^{-1} between dimethylamine and its radical cation.[13] The change of Δ is -0.37 to $+0.04 \text{ eV}$ over ε and r ; it is negative because the stronger attraction of the pair outweighs the desolvation of one ion, and it is the most favourable case for a small Δ that the continuum picture allows.

Two-ion. Both ions are moved with the density-functional continuum protocol of the previous section, using the separated-ion energies at $\varepsilon = 4$ and 10 (Table S10) against those at $\varepsilon = 78.4$ (Table S9), for methylamine and benzylamine with lumiflavin. The change of Δ , including the Coulomb term, is $+0.01$ to $+0.56 \text{ eV}$.

Substrate-induced shift. Without substrate the oxidized/semiquinone and semiquinone/reduced midpoints of the MAO A flavin are -159 and -262 mV , [15] so the two-electron midpoint is -211 mV . With substrate the full-reduction midpoint shifts by up to $+0.5 \text{ V}$ and no semiquinone is observed.[15] If the absence of semiquinone reflects equilibrium, the first one-electron potential lies at or below the two-electron midpoint, at most $-211 + 500 \text{ mV} = +0.29 \text{ V}$, a shift of the one-electron couple of at most 0.45 V . The main text uses the full $+0.5 \text{ V}$, which lowers Δ more than this reading allows. All three estimates are computed in `qbscreen/make_numbers_v6.py`.

S11 Cryptochrome

Orientation-averaged, perpendicular and parallel response of the cryptochrome pair over the exchange prefactor J_0 (both signs of J_{EH} ; $J = -2J_{\text{EH}}$, $J_{\text{EH}} = J_0 e^{-\beta r}$, $\beta = 14 \text{ nm}^{-1}$, $r = 1.90 \text{ nm}$ [16]), $k_S = k_P = 1 \text{ }\mu\text{s}^{-1}$, coherent (Table S11).

References

- [1] R. H. Bartels and G. W. Stewart. Solution of the matrix equation $AX + XB = C$. *Commun. ACM*, 15:820–826, 1972.
- [2] Q. Sun et al. Recent developments in the PySCF program package. *J. Chem. Phys.*, 153:024109, 2020.

Table S8: Coupling scan. “valid”: non-clashing geometries of those computed.

amine	approach	r_{N5N} (Å)	valid	median $ t $ (meV)	max $ t $ (meV)	min contact (Å)
methylamine	face	3.0	5/5	184.6	311.6	2.32
methylamine	face	3.5	5/5	52.8	105.6	2.72
methylamine	face	4.0	5/5	24.6	37.9	3.12
methylamine	face	4.5	5/5	5.1	31.1	3.53
methylamine	face	5.0	5/5	1.9	14.7	3.96
methylamine	face	5.5	5/5	1.0	4.2	4.40
methylamine	face	6.0	5/5	0.3	1.1	4.85
methylamine	edge	3.0	0/3	—	—	0.77
methylamine	edge	3.5	0/3	—	—	0.93
methylamine	edge	4.0	0/3	—	—	1.29
methylamine	edge	4.5	2/3	97.0	170.8	1.71
methylamine	edge	5.0	3/3	23.1	81.0	2.17
methylamine	edge	5.5	3/3	8.6	35.0	2.65
methylamine	edge	6.0	3/3	3.2	12.7	3.13
benzylamine	face	3.0	0/5	—	—	0.61
benzylamine	face	3.5	2/5	47.1	52.3	0.89
benzylamine	face	4.0	2/5	5.2	7.9	1.31
benzylamine	face	4.5	3/5	6.1	35.2	1.77
benzylamine	face	5.0	5/5	4.2	21.1	2.25
benzylamine	face	5.5	5/5	1.0	8.3	2.73
benzylamine	face	6.0	5/5	0.9	2.0	3.22
benzylamine	edge	3.0	0/3	—	—	1.00
benzylamine	edge	3.5	0/3	—	—	1.00
benzylamine	edge	4.0	0/3	—	—	0.92
benzylamine	edge	4.5	0/3	—	—	0.82
benzylamine	edge	5.0	0/3	—	—	1.00
benzylamine	edge	5.5	0/3	—	—	1.36
benzylamine	edge	6.0	1/3	0.4	0.4	1.78

Table S9: One-electron potentials in water.

couple	DFT (V vs NHE)	experiment	error (V)
$\text{Me}_2\text{NH}^{\bullet+}/\text{Me}_2\text{NH}$	+0.73	+1.300	-0.57
$\text{MeNH}_2^{\bullet+}/\text{MeNH}_2$	+1.26	—	—
$\text{BnNH}_2^{\bullet+}/\text{BnNH}_2$	+1.45	—	—
$\text{Fl}/\text{Fl}^{\bullet-}$ (exp.: FMN, pH 7)	-1.16	-0.313	-0.84

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Table S10: Raw DFT driving force of the contact pair.

continuum	amine	separated (eV)	$r = 4 \text{ \AA}$	$5 \text{ \AA}$	$6 \text{ \AA}$
$\varepsilon = 4$	methylaniline	3.55	2.65	2.83	2.95
$\varepsilon = 4$	benzylaniline	3.59	2.69	2.87	2.99
$\varepsilon = 10$	methylaniline	2.83	2.47	2.55	2.59
$\varepsilon = 10$	benzylaniline	2.97	2.61	2.69	2.73

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Table S11: Cryptochrome response (%).

J_0 (μT)	J (MHz)	powder (%)	$\perp$ (%)	$\parallel$ (%)
8.00e12	-1.26	+0.417	+0.494	+0.027
8.00e12	+1.26	+0.386	+0.464	-0.023
1.01e13	-1.58	+0.395	+0.469	+0.015
1.01e13	+1.58	+0.384	+0.457	-0.030
1.27e13	-1.99	+0.371	+0.454	-0.005
1.27e13	+1.99	+0.392	+0.461	-0.034
1.60e13	-2.51	+0.403	+0.498	+0.005
1.60e13	+2.51	+0.413	+0.491	-0.029
2.01e13	-3.16	+0.470	+0.561	+0.021
2.01e13	+3.16	+0.435	+0.536	-0.020
2.53e13	-3.98	+0.573	+0.681	+0.027
2.53e13	+3.98	+0.444	+0.580	-0.012
3.18e13	-5.01	+0.634	+0.751	+0.042
3.18e13	+5.01	+0.404	+0.483	+0.007
4.01e13	-6.30	+0.559	+0.622	+0.034
4.01e13	+6.30	+0.334	+0.407	+0.025
5.05e13	-7.93	+0.383	+0.420	+0.061
5.05e13	+7.93	+0.242	+0.320	-0.005
6.35e13	-9.99	+0.441	+0.534	+0.033
6.35e13	+9.99	+0.242	+0.299	+0.049
8.00e13	-12.57	+0.492	+0.589	+0.015
8.00e13	+12.57	+0.168	+0.234	+0.021
1.01e14	-15.83	+0.282	+0.372	+0.024
1.01e14	+15.83	+0.153	+0.174	+0.081
1.27e14	-19.93	+0.085	+0.115	-0.060
1.27e14	+19.93	+0.150	+0.131	+0.126
1.60e14	-25.09	+0.061	+0.078	+0.020
1.60e14	+25.09	+0.185	+0.119	+0.214
2.01e14	-31.58	+0.044	+0.049	+0.023
2.01e14	+31.58	+0.055	+0.078	+0.063
2.53e14	-39.76	-0.009	-0.004	-0.013
2.53e14	+39.76	+0.036	+0.041	+0.019
3.18e14	-50.05	+0.015	+0.017	+0.009
3.18e14	+50.05	+0.068	+0.063	+0.041
4.01e14	-63.01	+0.010	+0.011	+0.007
4.01e14	+63.01	+0.052	+0.045	+0.031
5.05e14	-79.33	+0.004	+0.005	+0.003
5.05e14	+79.33	+0.039	+0.049	+0.015
6.35e14	-99.87	+0.003	+0.003	+0.002
6.35e14	+99.87	+0.034	+0.036	+0.019
8.00e14	-125.73	+0.001	+0.002	+0.001
8.00e14	+125.73	+0.011	+0.017	+0.002